

ForLoop Control Panel

SciLoop AI Panel

A clear, simple explanation of how the SciLoop AI panel was developed, how it connects to SciLoop AI, and which backend APIs power the experience.

[Control Panel](#)[Secure Backend APIs](#)[AI Explanation](#)[News to Visual Plan](#)[Provider Switching](#)

What We Created

- A SciLoop AI panel inside the ForLoop Control Panel.
- A backend-safe place to check AI server status, readiness, and provider keys.
- A control layer that connects SciLoop News Portal, AI explanation, simulation, and Visual Language Lab.
- A secure route where API keys stay server-side instead of being exposed in the browser.
- A demo-ready dashboard that makes the AI system visible and controllable.

3001

ForLoop Control API

5050

SciLoop AI Backend

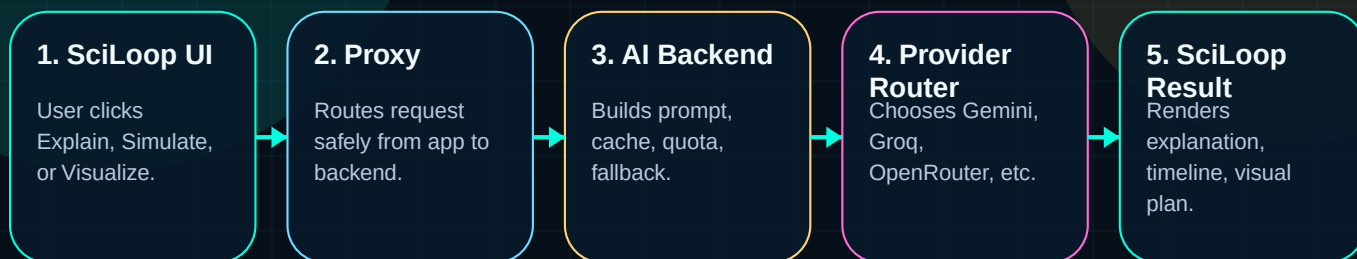
Secure

Keys stay backend-side

Demo line: I built the SciLoop AI control layer that connects the frontend, ForLoop dashboard, backend, provider APIs, and visual explanation system.

How the SciLoop AI Panel Connects Everything

The panel works like an operator dashboard. SciLoop is the user-facing product, ForLoop is the control layer, and sciloop-backend is the secure AI service.



ForLoop Control Panel Role

Shows SciLoop AI status, saves/checks provider keys, starts the allowlisted backend target, and logs admin actions for demo visibility.

SciLoop AI Backend Role

Receives article/concept data, builds prompts, calls providers server-side, handles fallback, and returns clean JSON to the UI.

Important: API keys are not placed inside the HTML page. The browser talks to the backend; the backend talks to AI providers.

Main APIs Behind the SciLoop AI Panel

These are the important routes used by the control panel and the SciLoop AI experience.

GET /api/admin/sciloop-ai/status	Checks backend/provider visibility for the ForLoop panel.
GET /api/admin/sciloop-ai/readiness	Shows if enough provider keys are ready for a live demo.
POST /api/admin/sciloop-ai/keys	Saves provider keys locally in backend env files.
POST /api/admin/sciloop-ai/check-provider	Tests one provider and returns ready, missing key, quota, or warning.
POST /api/admin/sciloop-ai/check-all	Runs full provider readiness check.
POST /api/admin/sciloop-ai/start-server	Starts the SciLoop AI backend target safely.
POST /api/sciloop-ai/explain	Turns news into simple meaning, timeline, people/team info, impact, and visual blueprint.
POST /api/sciloop-ai/simulate	Creates a possibility simulation storyline.
POST /api/sciloop-ai/news-visualize	Creates visual-plan handoff for Visual Language Lab.
GET /api/sciloop-ai/news-visualize/:handoffId	Loads the stored visual-plan handoff after redirect.

Provider APIs Used

Gemini for structured explanation and planning. Groq for fast responses. OpenRouter, Hugging Face, Together and others as optional routes. Local fallback keeps demos alive.

What Happens When a User Clicks Explain

1 User Action

User opens News Portal and clicks Explain with SciLoop AI.

2 Payload

Frontend sends title, summary, source, URL, and date.

3 Prompt

Backend builds a structured prompt for simple explanation.

4 Provider

Provider router uses available APIs and switches if one fails.

5 JSON

Backend returns explanation, timeline, people/team info, impact, and blueprint.

6 Live UI

SciLoop shows colorful explanation with typing effect and simulation action.

What the User Sees

Simple meaning, why it matters, people/team if confirmed, animated timeline, visual blueprint, and Simulate button.

What the Backend Handles

Prompt construction, provider selection, quota/cache protection, error handling, and fallback explanation.

Demo sentence: One click sends the article through a secure backend, talks to an AI provider, and returns a student-friendly SciLoop explanation.

Why This Architecture Is Safe and Demo-Ready

Keys stay hidden

Provider keys are read from backend environment files. The browser never receives real API keys.

Controlled panel

ForLoop admin actions use an access code and allowlisted runtime targets.

Failure-safe UI

If a provider fails, hits quota, or times out, SciLoop can switch provider or use local fallback.

Quota protection

Cache, cooldowns, and provider status checks protect limits during demos.

Clean JSON

The UI receives predictable fields for explanation, timeline, simulation, and visual plans.

Expandable

The same pattern can support more providers, subjects, voice, visuals, and future learning tools.

How to Explain: I Created SciLoop AI

- I created the SciLoop AI experience layer: explanation panel, timeline, simulation button, and visual handoff.
- I connected the frontend to a secure backend instead of exposing API keys.
- I built the ForLoop SciLoop AI panel for readiness, provider checks, backend status, and runtime start.
- I connected News Portal to AI explanation and Visual Language Lab.

Simple Demo Script

Opening

This is the ForLoop Control Panel. I built a SciLoop AI panel inside it so I can check and control the AI layer of my SciLoop platform from one place.

Backend

The AI keys are not inside the website. They stay in the backend. The frontend only sends safe requests, and the backend talks to providers.

News Portal

When a user clicks Explain with SciLoop AI, the article title and summary go to my backend, and the backend returns structured learning content.

Visualize

When the user clicks Visualize, SciLoop creates a visual plan and opens the Visual Language Lab so the idea becomes a causal visual scene.

Final One-Line Summary

ForLoop controls the AI system, SciLoop AI explains and simulates the news, and the Visual Language Lab turns that explanation into a visual learning scene.

Best demo order: ForLoop Control Panel -> check SciLoop AI status -> open SciLoop News Portal -> Explain -> show live explanation -> Simulate or Visualize.